Plasma Chemistry and Plasma Processing (2026) 46:27
https://doi.org/10.1007/s11090-026-10648-6

ORIGINAL PAPER

Using Viral Aerosol Fluorescence for Detection of Virus Infectivity Change Induced by Non-thermal Plasma

Zhenyu Ma[1] · **Herek L. Clack**[1]

Received: 14 August 2025 / Accepted: 29 January 2026 / Published online: 11 February 2026

Abstract

Airborne transmission of infectious diseases poses a great threat to public health and the global economy, prompting increased interest in the detection and mitigation of infectious airborne pathogens. Development of air disinfection technologies – those that rely on pathogen neutralization or inactivation rather than particle filtration, such as non-thermal plasma (NTP) – can require extensive tests to determine how the degree of disinfection is affected by operational settings, environmental conditions, aerosol composition, pathogen type, among other factors. Such parametric evaluation is made much more tedious by the need to physically extract pre- and post-treatment aerosol samples for comparative microbiological assays in order to measure change in pathogen viability and the efficacy of the neutralization/inactivation process. UV aerosol fluorescence has been used to detect and characterize aerosols of biological origin, such as pollen. In this study, UV fluorescence of MS2 bacteriophage aerosol is studied for its potential to serve as an indicator of changing viral aerosol infectivity, one that can provide a quicker indication of a change in viral aerosol infectivity than conventional bioaerosol collection followed by microbiological assay. In the present study, infectivity assays and fluorescence measurements of viral aerosols are taken before and after non-thermal plasma treatment. Results indicate that NTP treatment induces infectivity loss and diminished fluorescence intensity. Diminished fluorescence intensity and reduced infectivity are positively correlated, both becoming more pronounced with increased intensity of non-thermal plasma treatment. These findings suggest UV aerosol fluorescence could serve as a fast indicator of airborne virus infectivity change during test, evaluation, and optimization of air disinfection processes like NTP.

Keywords Virus · Fluorescence · Aerosol · Infectivity · Non-thermal plasma

✉ Zhenyu Ma
zhenyuma@umich.edu

Herek L. Clack
hclack@umich.edu

[1] Department of Civil and Environmental Engineering, University of Michigan, 1351 Beal Ave, Ann Arbor, MI 48109, USA

Introduction

Airborne transmission of infectious diseases threatens wildlife (e.g., avian influenza), livestock (e.g., foot and mouth disease), and humans alike [1]. Viral pathogens, in particular, can mutate as they exchange genomic traits of different host species (e.g., zoonotic transmission between animals and humans of H5N1 influenza [2, 3]) or resulting from host co-infections with other viral diseases (e.g., SARS-CoV-2 and influenza [4]), reducing the durability of host immunity conferred from either vaccination or prior infection. Furthermore, for emerging infectious diseases without effective vaccines, an airborne route of transmission can lead to rapid disease spread within a susceptible population if rapid testing is unavailable or if population isolation measures (e.g., lockdowns) are considered untenable.

Throughout history, accepted theories of infectious disease transmission underwent dramatic shifts (see, for example, the historical review of Jimenez et al. [5]). Without sustained support for scientific investigation and engineering innovation, advances in the study of infectious aerosols, including methods development for their collection, preservation, and characterization, have lagged. Despite the recent emergence of some novel methods for infectious aerosol aging [6] and collection [7], most methods in use are based on more classical approaches [8]. Such methods are critical to understanding how infectious aerosols become less infectious - either through natural decay in the ambient atmosphere or as a result of engineered mitigation measures for disinfecting air. Air disinfection reduces exposure of the individual to infectious aerosols, thereby reducing infection risk. In contrast, as a strategy, population vaccination boosts an individual's immunity to infection once exposed to infectious aerosols. Methods of air disinfection have historically targeted the physical characteristics of infectious aerosols, achieving air disinfection by using physical filtration to strip particulate matter from air, infectious aerosols being a sub-classification of particulate matter [9, 10]. More recently, however, further shifts in infectious disease transmission theory, along with growing awareness of the practical limitations of particle filtration methods [11, 12], have spurred interest in air disinfection methods that target the infectivity, rather than the aerosol characteristics, of infectious aerosols.

Among the air disinfection technologies that target aerosol infectivity are non-thermal plasmas (NTPs) directly applied to flowing airstreams. Non-thermal plasmas have been applied to treat pathogenic organisms, including bacterial [13], fungal [14], and viral aerosols [15, 16, 17]. Compared to other methods of disinfecting flowing airstreams, NTPs have been shown [18, 19] to achieve comparable levels (97–99%) of viral aerosol disinfection as HEPA filtration [20] and UV irradiation [21], with the added advantages of 4–5 times greater treated air flow rate, 25% lower pressure drop than HEPA filtration, and up to 44 times smaller treatment volume and flow area than either UV or HEPA filtration. Comparing the performance of UV irradiation, HEPA filtration, and direct application of NTP is straightforward since all three operate as steady flow processes where pre- and post-treatment conditions and pre- and post-treatment aerosol samples are both well-defined. By contrast, NTPs that generate and release disinfectant species into indoor spaces operate on fumigation principles where species concentrations vary in space and time, greatly complicating measurement of performance or efficacy. For these reasons, the present study explicitly excludes from consideration plasma applications that are dispersive or fumigating in nature.

Measuring the performance of air disinfection technologies is burdensome, especially when targeting airborne viruses. Traditionally, virus infectivity is assessed by observing the impacts of a virus on its host organism, either qualitatively (cytopathic effects [22]) or quantitatively (plaque assay [23]). Presently, performance measurements of air disinfection technologies involving viral aerosols require collecting paired inlet/outlet aerosol samples extracted from the airflow undergoing disinfection. These viral aerosol samples are then subjected to plaque assay, a time- and labor-intensive process that involves inoculating the collected viral aerosol samples onto multiple colonies of the susceptible host organism, prepared in advance, followed by incubation and plaque enumeration processes. Depending on the host organism, the growth/cultivation period after virus inoculation can range from hours (e.g., bacteriophage MS2) to days (e.g., MDCK cells) before plaque assays can be completed and the results of a single air disinfection trial can be determined [24, 25]. Conducting parametric performance testing of air disinfection technologies in this way is both time-consuming and resource-intensive, burdens that increase as testing advances from prototype or pilot scale to full-scale devices with higher air flow rates. Given these considerations, a more rapid, less laborious method for measuring the performance of air disinfection technologies is needed.

Fluorescence spectroscopy has been used to detect aerosols of biological origin, such as pollen [26] or allergens with a protein structure. Originally developed for rapid detection of biowarfare agents, ultraviolet laser-induced fluorescence (UV-LIF) is suitable for any biological particles that have macromolecular fluorophores, including but not limited to proteins that contain tryptophan amino acid, NAD(P)H enzymes, or riboflavin [27, 28]. Those biological fluorophores have intrinsic fluorescence that can be detected by particle fluorescence spectroscopy in real-time to provide information on their chemical composition, usually within hours, if not minutes [29]. The intrinsic fluorescence inherent in bioaerosols due to their protein content also presents the potential for real-time analysis of the effects of disinfection treatments on airborne viruses. For direct non-thermal plasma exposure, the short-lived oxidative species are argued to play an important role in the NTP inactivation process of airborne pathogens [19]. The commercialization of UV aerosol fluorescence spectrometer instruments like the Wideband Integrated Bioaerosol Sensor (WIBS) and the ultraviolet aerodynamic particle sizer (UVAPS) has enabled an increasing number of studies whose objective is to document bioaerosol abundance in ambient air under different contexts. The present study proposes to extend the use of such instruments from ambient bioaerosol detection to use as a real-time diagnostic indicator of infectious aerosol viability before and after an air disinfection process; in the present study, the chosen process is direct NTP exposure for inactivation of viral aerosols.

In the present study, the measured viral aerosol fluorescence is found to be correlated with the degree of inactivation by direct NTP exposure. In other words, the change in viral aerosol fluorescence and the degree of viral aerosol inactivation are found to correlate with each other over a range of NTP settings.

Materials and Methods

General Test Apparatus Setup and Experimental Procedure

In this study, MS2 bacteriophage is used as a surrogate for airborne infectious viruses such as influenza, respiratory syncytial virus (RSV), and coronavirus (SARS-CoV-2), chosen because it is relatively easy to propagate to high liquid-phase titers that, once nebulized into a flowing airstream, are diluted down to much lower concentrations (viruses per volume of air) in the aerosol phase. As a biosafety level 1 (BSL-1) microorganism, intentionally aerosolizing MS2 for airborne disinfection experiments can be done safely, as compared to strict prohibitions against generating aerosols when working with BSL-2 (and higher) microorganisms. The experimental setup, as shown in a previous paper (Fig. 2.2 from Ma et al. [17]), consists of a virus aerosol generator, aerosol drying section, the plasma reactor, sampling ports, and an induced-draft fan. A nebulizer (Mesh Nebulizer YM-3R9) is used to generate MS2 viral aerosols at an aerosolization rate of 0.3–0.5 mL per minute. After passing through a desiccation section, the MS2 aerosol passes through a non-thermal plasma (NTP) reactor before being directed into an exhaust vent by an in-line induced-draft fan (Fantech FGC-4). Sampling ports for sampling viral aerosols immediately before and after they are exposed to plasma are located both upstream and downstream of the NTP reactor.

During a single experimental trial lasting 30 min, a total of 10 mL of MS2 virus stock is aerosolized into an airflow of approximately 200 L/min passing through the NTP reactor. Extractive sampling of the airflow at 1.5 L/min through sampling ports upstream and downstream of the plasma reactor collects aerosol samples into twin impingers (Ace Glass Model 7533-18), where viral infectivity is preserved for subsequent virus infectivity assay. A vacuum pump (GAST Model DOA-P704-AA) and twin flow regulators (Omega FLDA3214G) control the draw of aerosol-laden air through the impingers, each of which contains 15 mL of impinger liquid (sterilized 1X phosphate-buffered saline, with the addition of 0.5 mmol/L of sodium sulfide). At the end of each trial, the liquid from each impinger and the viral aerosols collected within it are refrigerated at 4 °C before virus infectivity assays. For aerosol fluorescence measurements, a UV aerosol fluorescence spectrometer (wideband integrated bioaerosol sensor (WIBS), Droplet Measurement Technologies, Longmont, Colorado) extractively samples viral aerosols from the same upstream and downstream sampling ports for 1 min. For all trials, a plasma-off control trial is performed while the plasma is unpowered.

Non-Thermal Plasma Reactor Setup

The device used in this experiment to achieve virus inactivation is a packed-bed dielectric barrier discharge (DBD) non-thermal plasma reactor, whose detailed configuration has been described in previous literature [17]. To summarize, the reactor itself comprises two smaller Plexiglass tubes (OD of 3.5 inches, length of 12 inches) with rubber rings located 1 inch from one end, sliding freely in a larger Plexiglass tube (OD of 4 inches, length of 8 inches) that connects those two. On each end of the smaller tube that is located inside the larger one, a brass perforated plate serves as the electrode, with 500 glass beads (McMaster-Carr 8996K25) packed in between the two brass electrodes. A brass ring is attached around the larger tube on the outside, located at the packed-bed glass beads section, serving as the positive electrode for the reactor. In operation, the induced draft fan at the downstream end pulls virus-containing air into the connected tube sections at a flow rate of around 200 L per minute (lpm), then the viral aerosols pass through the perforated brass electrodes and glass beads packed-bed section, where plasma discharge is formed, to be inactivated. Plasma power is controlled by a wave function generator (BK Precision, Model 4052) and an amplifier (Trek Model-610E). In this study, the wave function generator outputs an alternating current (AC) sinusoidal wave with a frequency of 300 Hz, and then the amplifier amplifies the peak-to-peak voltage from 20 Volts to 20 kilovolts as the plasma power source.

Detection of Aerosol Fluorescence Via a Wideband Integrated Bioaerosol Sensor (WIBS)

The Wideband Integrated Bioaerosol Sensor (WIBS) is used to measure the virus aerosol fluorescence. WIBS is a single aerosol particle fluorescence monitor that measures the size, fluorescence, and other properties of the detected aerosol particles using light scattering and fluorescence spectroscopy [29, 30]. The instrument draws in an aerosol-laden air stream at a rate of 0.3 L/min surrounded by a sheath flow of 2.1 L/min via a laminar-flow delivery system. Aerosols first pass through a particle detection system comprised of a Class 3B 635 nm diode laser that can detect individual particles from 0.5 µm to 50 µm, up to 120 per second. Particles that exceed a user-selectable laser scattering intensity correlated to particle size will trigger two UV excitation (280 nm and 370 nm) Xenon flash lamps in close succession. Induced particle fluorescence is measured by two photomultiplier tubes that detect fluorescence emission in wavelengths ranging from 310 to 400 nm and 420–650 nm. These wavelengths are controlled via optical components within the WIBS instrument and considered optimal for the detection of natural fluorophores in general [29, 30]. Similar types of commercialized UV-fluorescence detection instruments have also adopted comparable wavelength ranges [31, 32]. Both fluorescence emission bands can be measured in response to either of the excitation wavelengths, except for 370 nm excitation and 310–400 nm emission, because the elastically scattered 370 nm Xenon lamp energy would saturate the 310–400 nm detector. Other particle-related parameters, such as asphericity, are also measured by the instrument via other methods but will not be described in detail here, as this study only uses the size and fluorescence data of the detected particles.

Particle fluorescence data is obtained on a particle-by-particle basis, which is summarized and presented for tens of thousands of particles in the Results section. For each individual result plotted as a distribution curve in a figure or reported as a numerical value of the median fluorescence intensity, the original data comprise at least 2,000 (and up to 10,000) independent measurements of single-particle fluorescence intensity. In this study, particle fluorescence signals within the 310–400 nm wavelength band resulting from an excitation wavelength of 280 nm are analyzed in detail.

Virus Preparation and Infectivity Assay

MS2 virus is used as the surrogate virus throughout this study. The original ssRNA Bacteriophage MS2 (ATCC 15597-B1) and its corresponding host *E. coli* ATCC 15597 were purchased from American Type Culture Collection (ATCC). Following established virus propagation protocols [33] and sterilization by passing through a 0.22 µm polyethersulfone filter, the MS2 stock has a virus titer of approximately 5 × 10^{12} plaque-forming units (pfu) per mL.

For virus infectivity tests and viral aerosol fluorescence analysis, the original virus stock is diluted 20 times with phosphate-buffered saline (PBS) into a total volume of 10 mL (0.5 mL of stock). Then the virus is aerosolized, exposed to NTP, and collected into impingers following the aforementioned procedures. Plaque assays are performed to test the infectivity of collected viral aerosols and enumerate the viable viruses, following a standard double-agar-layer protocol [24]. Firstly, serial dilutions of the original collected sample are made by diluting the sample with PBS buffer to a desired concentration range. Then 100 µL of the diluted virus sample is added to 3 mL of top layer agar media, along with 100 µL of host bacteria media. The virus-host bacteria mixture is poured onto a pre-made hard agar plate and incubated overnight at 37 °C, followed by an enumeration of the number of plaques on the plate. Based on the calculated plaque-forming units concentration (pfu/mL) from plates, the number of viable viruses in the original sample is subsequently back-calculated. The loss of infectivity, in units of log pfu/mL, is calculated based on the difference in viable virus concentrations determined from aerosol samples collected upstream and downstream of the NTP reactor during plasma-on (trial) conditions and plasma-off (control) conditions. For each individual trial, duplicates of virus samples, at a minimum, were collected and analyzed. For each type of experiment (for example, a given plasma peak-to-peak voltage and frequency), at least three individual trials were performed.

Results and Discussion

Characteristic Changes in MS2 Viral Aerosol Fluorescence Due to Exposure to Non-thermal Plasma

In the present study, only WIBS measurements of aerosol fluorescence intensity within the 310–400 nm emission wavelength band (280 nm excitation wavelength) are considered; however, at an aerosol sampling rate of up to 120 Hz and a sampling period of 1 min for each trial or control condition, the presented data are generally ensembles of thousands of aerosol fluorescence measurements.

In Fig. 1, the black trace represents electronic noise within the instrument, produced by manually triggering the Xe lamp with the detection chamber empty, and represents the response of the detection chamber, photodetector, and associated electronics to the Xe lamp excitation (280 nm) in the absence of particles [30]. Even in the absence of particles, stochasticity within the detection system causes this forced-trigger background noise to take the form of a fluorescence intensity distribution; a similarly shaped distribution of higher fluorescence intensity represents measurements from MS2-containing aerosols (Fig. 1, blue trace). The fluorescence intensity distribution for MS2-containing aerosols lies to the right of the background distribution, which is indicative of the viral aerosol fluorescence being distinguishable from statistical and measurement noise.

Figure 2 presents the raw data from Fig. 1 in the form of a histogram of aerosol fluorescence intensity measurements. As the histogram suggests, the fluorescence intensity of individual MS2 aerosols can vary in a wide range. Therefore, the number of datapoints (sample size) is important to the accurate representation of the fluorescence intensity distribution

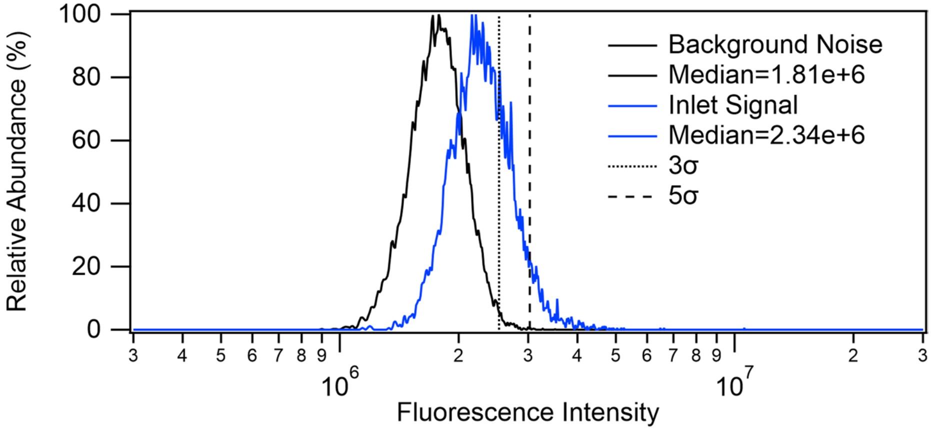

Fig. 1 Background electronic noise (black) and fluorescence intensity distribution for 4607 particles produced from aerosolizing MS2 stock (blue). Fluorescence excitation wavelength: 280 nm; emission wavelength 310–400 nm. The 3σ and 5σ dashed lines represent the values for three times and five times the standard deviation of measured background fluorescence intensity

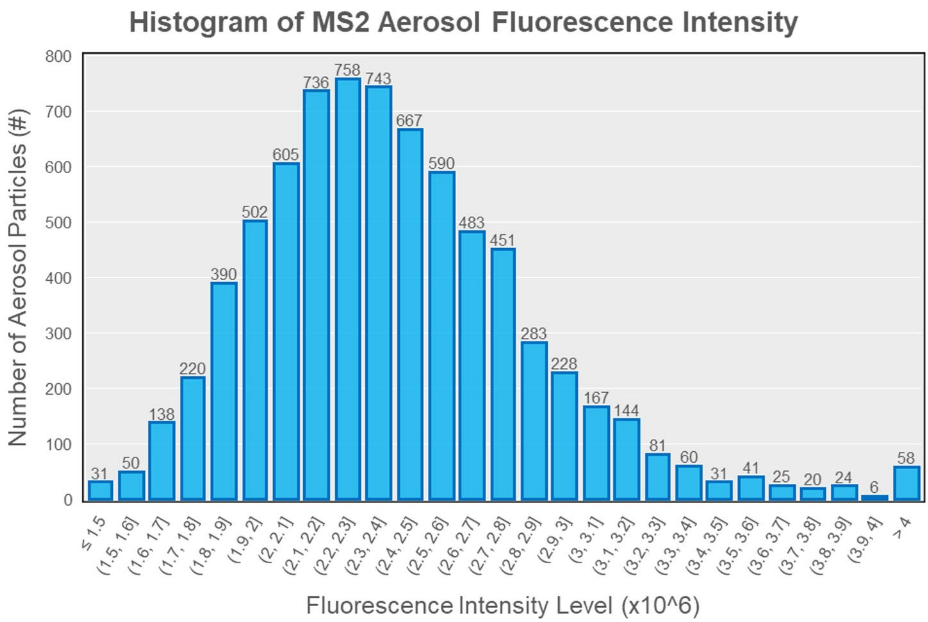

Fig. 2 Histogram of WIBS fluorescence intensity measurements of individual MS2 aerosol. The number on top of each column represents the number of particles within the given fluorescence intensity range

of aerosols sampled by the WIBS instrument. For this reason, fluorescence intensity distributions are analyzed throughout this study, with each comprised of over 5,000 individual aerosol fluorescence measurements.

Figure 3 is an example of how exposure to non-thermal plasma changes viral aerosol fluorescence intensity. Using the experimental approach described in *Materials and Methods*, aerosols (typically ~ 10^4 in number) sampled after plasma exposure (20 kV applied peak-to-peak voltage) exhibit a fluorescence intensity distribution that is collectively reduced (shifted leftward) as compared to the fluorescence intensity distribution of aerosols sampled before plasma exposure (Fig. 3, upper). By comparison, a much smaller shift occurs in aerosols sampled from the same locations when no plasma is generated in the NTP reactor (0 kV) (Fig. 3, lower). Quantitatively, when no plasma is generated in the NTP reactor (0 kV), the median aerosol fluorescence intensity decreased by 4% from inlet to outlet (from 2.34×10^6 to 2.25×10^6). In contrast, when plasma is generated (20 kV), the median aerosol fluorescence intensity decreased by 11% from 2.34×10^6 to 2.09×10^6, with a p-value of

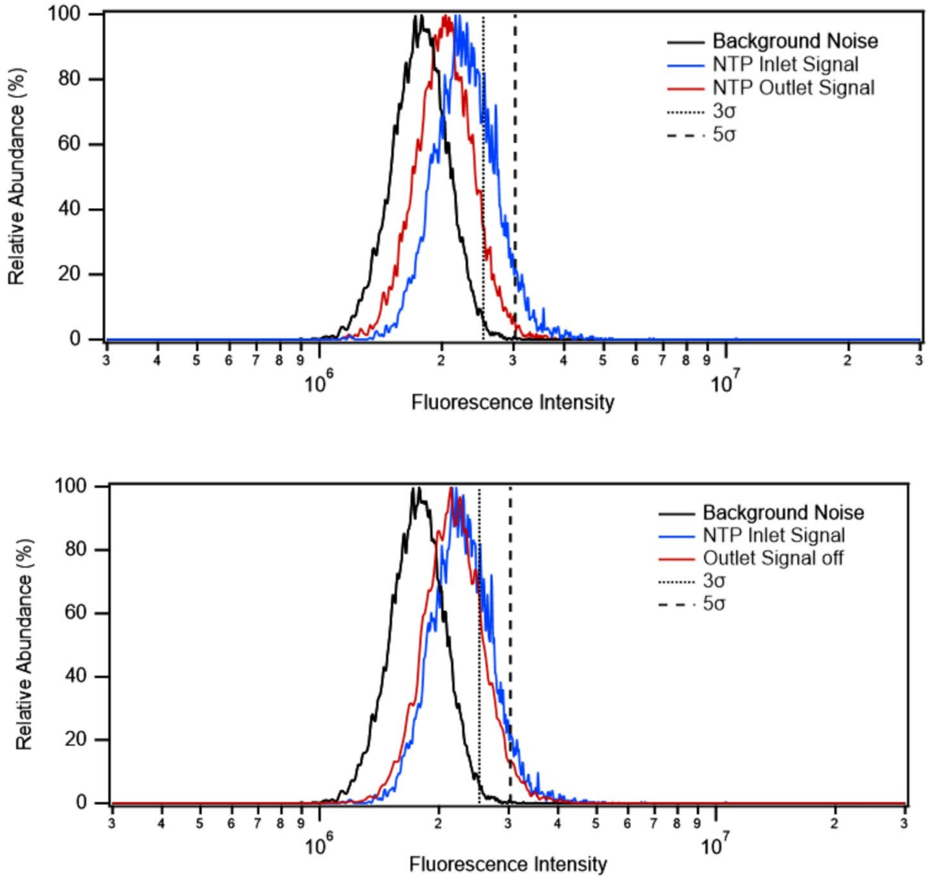

Fig. 3 Fluorescence intensity distributions of particles produced from aerosolizing MS2 stock and sampled at the inlet (blue) and outlet (red) of a powered (upper) and unpowered (lower) non-thermal plasma reactor. Plasma operating conditions: 20 kV plasma peak-to-peak voltage, 300 Hz driving frequency, 1.4 Watts plasma discharge power, 0.3 s exposure time

7.9×10^{-24} for a two-tailed t-test, suggesting a statistically significant difference between the two groups of measured fluorescence. In addition, all measurements involving viral aerosols have a median fluorescence significantly higher than the background level (1.82×10^6). Using relative (Fig. 3) rather than absolute (Fig. 2) distributions reduces the potential for sample size bias because of aerosol number concentration or aerosol size distribution changes between the NTP reactor inlet and outlet. Previous measurements of aerosol number concentration (via paired condensation particle counters) and aerosol size distribution (via scanning mobility particle sizer) have found [17] that aerosol size distributions are largely unchanged between reactor inlet and outlet under both powered and unpowered conditions. Aerosol number concentration at the non-thermal plasma reactor outlet was approximately 26% lower than at the inlet (70000/cm^3 vs. 95000/cm^3) at 0 kV and 37% lower (60000/cm^3 vs. 95000/cm^3) at 20 kV. However, irrespective of sampling location or applied voltage, aerosol sample sizes that comprise WIBS-measured fluorescence intensity distributions are largely constant (~ 10^4). These results suggest that changes in aerosol size or concentration are unlikely to be responsible for the observed changes in aerosol fluorescence intensity. Most importantly, whatever changes in size or concentration may occur, neither has the potential to induce a significant lowering of the median fluorescence intensity, which should remain statistically highly representative at sample sizes of around 10^4.

Deeper analysis of measured fluorescence intensity data reveals that the dilute starting suspension of bacteriophage MS2 in 1X phosphate buffered saline (PBS) (8 g/L NaCl, 0.2 g/L KCl, 1.15 g/L Na_2HPO_4, 0.2 g/L KH_2PO_4), once nebulized and injected into an airstream, dries into polydisperse aerosols characterized by a large number of small, weakly fluorescent or non-fluorescent particles constituting a large fraction of the overall aerosol number concentration (#/cm^3). Fluorescence data obtained for particles produced by nebulizing PBS solution alone show more than 90% of particles measured less than 1 μm in diameter (Table 1). Recorded fluorescence intensity distributions of these particles displayed poor separation from the background fluorescence trace (+/- 3σ) associated with instrument background noise. Scatterplots of particle fluorescence intensity versus particle size for all particles produced from nebulizing PBS (Figure SI-1) show that fluorescence intensity remains largely constant as particle size increases. In contrast, similar scatterplots of particles produced by nebulizing a suspension of MS2 phage (Figure SI-2) in PBS reveal gradually increasing fluorescence intensity with increasing particle size ($R^2 = 0.52$ for a 2nd-order polynomial fit). Figure 4 presents the same data using median fluorescence intensity values, showing

Table 1 Size distribution of inorganic particles from the aerosolized sample

Particle size (μm)	Count (#)	Frequency (%)
<=0.5	9991	1.697
0.5–0.6	258,617	43.925
0.6–0.7	147,006	24.968
0.7–0.8	59,984	10.188
0.8–0.9	34,400	5.843
0.9-1.0	23,895	4.058
1.0-1.1	16,921	2.874
1.1–1.2	11,938	2.028
1.2–1.3	7639	1.297
1.3–1.4	5155	0.876
1.4–1.5	3496	0.594
>1.5	9729	1.652

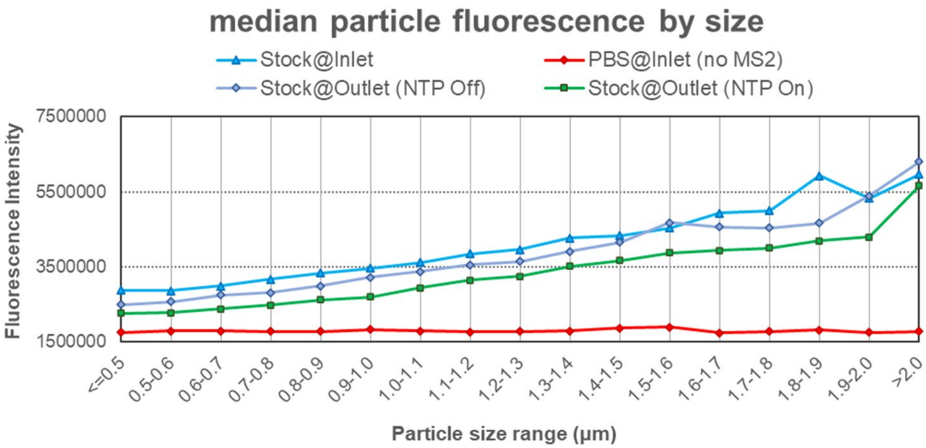

Fig. 4 Increasing median fluorescence intensity with particle size, for particles produced from nebulizing MS2 stock and sampled at the inlet (aqua) and outlet (green) of a powered non-thermal plasma reactor, and at the outlet of an unpowered reactor (lilac). Particles produced from aerosolizing PBS alone (red) exhibit an overall lower median fluorescence intensity and no increase in intensity with particle size

increasing median fluorescence intensity with increasing particle size for particles resulting from nebulizing MS2 phage in PBS, whereas median fluorescence intensity is constant as particle size increases for particles produced by nebulizing PBS alone. Even for particles resulting from the nebulization of MS2 phage in PBS, the smallest particles that form after drying of the smallest droplets have a greater chance of containing no MS2 virions. Depending on the pfu/mL of MS2 phage in PBS being nebulized, there will be a droplet diameter for which the droplet volume would contain less than 1 pfu. For particles formed from droplets that are smaller than this, it becomes a statistical probability in which one out of every *n* particles contains a single infectious virion. For example, if 10 mL of MS2 phage in PBS with a titer of 5×10^{12} pfu/mL, both typical values for the present study, were nebulized into 10^{13} monodispersed droplets of 1 µm diameter, each droplet on average would have a pfu count of 0.2, suggesting that only one in five droplets would contain a single virion. Fernandez et al. [6] demonstrated this experimentally by using droplet-on-demand dispensers to produce droplets from solutions of bacteria, fluorescent beads, and spores. Their results showed that spore, bacteria, and fluorescent bead counts per droplet varied linearly with initial solution concentration and fell below 1 per droplet when the initial solution concentration fell below a threshold value. Unfortunately, virions are not large enough to enumerate through microscopy, but using similar logic, nebulizing MS2 phage in PBS will produce particles that contain only the constituents of PBS and no MS2 virions. Because the smallest particles in a polydisperse aerosol often are overrepresented in their number concentrations even as they constitute only a tiny volume or mass fraction of all aerosols, such overrepresentation of particles containing not a single virion can suppress the fluorescence intensity that would otherwise be expected from particles resulting from nebulizing MS2 phage in PBS. For solutions of MS2 phage in PBS of decreasing pfu/mL, the fluorescence intensity distributions and median fluorescence intensity values of particles resulting from nebulization will both have a lower signal-to-noise ratio as the percentage of particles sampled by WIBS that do not contain a single MS2 virion grows.

In order to increase signal-to-noise ratio, provide greater dynamic range, and produce better separation between the fluorescence intensity distribution curves of the MS2 phage aerosols and the electronic noise of the instrument, we opted to limit further analysis to aerosols greater than 1 μm and kept the initial virus titer in the MS2 stock undiluted.

The relationship between particle size and fluorescence intensity informs the analysis of changes in viral aerosol fluorescence as a result of non-thermal plasma exposure. Figure 5 illustrates that the change in aerosol fluorescence intensity for particles resulting from nebulizing MS2 phage in PBS exposed to non-thermal plasma is more significant for those whose diameter is greater than 1 μm, as the median fluorescence intensity of particles dropped by 4.9×10^5 from NTP inlet to outlet when NTP is powered (from 4.36×10^6 to 3.87×10^6), whereas that only dropped by 2.5×10^5 for all particles (Fig. 3) (from 2.34×10^6 to 2.08×10^6). There is a relatively minimal difference between the aerosol fluorescence intensities of aerosols sampled upstream and downstream of the NTP reactor when the reactor is unpowered (bottom figure), but a much more pronounced decrease in aerosol fluorescence when the NTP is powered (top figure). The median particle fluorescence intensity at the NTP outlet was 11% lower than the inlet (3.87×10^6 compared to 4.36×10^6) for plasma-on and 4% lower for plasma-off (4.16×10^6 compared to 4.36×10^6). Although it is clear that 24–26 nm-sized [34] MS2 phage is small enough to fit within the 500 nm and larger-sized particles

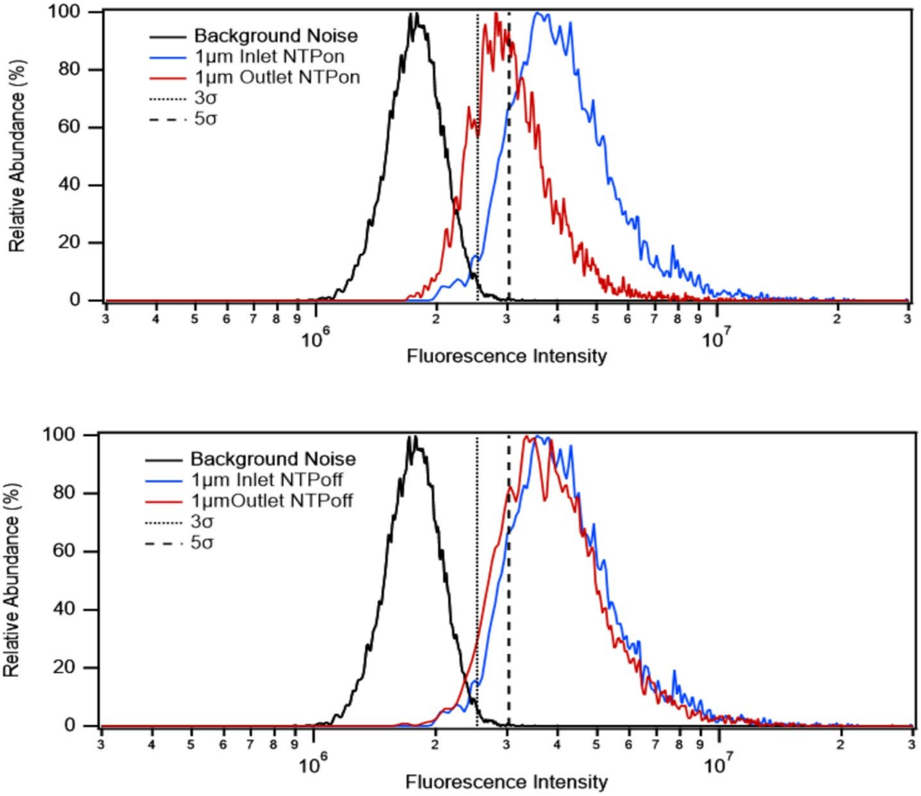

Fig. 5 Fluorescence intensity distributions of particles, larger than 1 μm, produced from aerosolizing MS2 stock and sampled at the inlet (blue) and outlet (red) of a powered (upper) and unpowered (lower) non-thermal plasma reactor

that can be sized by WIBS, comparing Fig. 5 to Fig. 3, it is apparent that particles larger than 1 µm (Fig. 5) exhibit more intense UV-induced fluorescence of MS2 phage and greater reduction of fluorescence intensity as a result of non-thermal plasma exposure than particles smaller than 1 µm. This likely reflects the greater probability for larger particles to contain one or more MS2 virions, as well as the greater probability for the smallest particles to contain no MS2 virions at all. It is noteworthy, however, that even under size-selective examination, weakly fluorescent sub-micron particles produced from nebulizing MS2 phage in PBS still show evidence of reduction in fluorescence intensity after non-thermal plasma exposure (Fig. 4) similar to the trend observed for larger particles (i.e., Intensity_inlet > Intensity_outlet_off > Intensity_outlet_on).

Correlation Between Viral Aerosol Fluorescence Intensity and Virus Infectivity

Plaque assays of MS2 phage aerosols were also performed before and after non-thermal plasma exposure, in parallel with WIBS-based UV fluorescence intensity measurements of the aerosols. A detailed discussion of NTP-induced viral aerosol inactivation and the scientific rationale for establishing such a strict definition that excludes particle removal processes is provided elsewhere [17]. Briefly summarizing, the degree of virus inactivation in the present study is calculated as the difference in collected viral aerosols' infectivity resulting from exposure of the aerosols to a disinfection process, excluding the effects of physical removal and experimental artifacts.

$$Inactivation\left(\frac{\log PFU}{mL}\right) = (C_{inlet,\,NTP\,on} - C_{outlet,\,NTP\,on}) - (C_{inlet,\,NTP\,off} - C_{outlet,\,NTP\,off})$$

Using the formula above for calculating the level of "NTP inactivation", physical aerosol removal processes, such as filtration, boundary losses, and electrostatic precipitation, are excluded from consideration as "inactivation" and are excluded from results reported as inactivation here.

Using the same experimental setup and conditions, a reduction in MS2 phage infectivity of 1.03-log was determined from plaque assays of MS2 phage aerosols collected upstream and downstream of the non-thermal plasma reactor. This level of inactivation does not represent a maximum for NTP technology and is typical of and specific to our experimental setup and conditions; other researchers have demonstrated both higher and lower levels of inactivation, as strictly defined above, employing different experimental setups, conditions, and viral aerosol targets [18, 19]. By sweeping non-thermal plasma discharge power, a correlative relationship becomes evident between MS2 phage inactivation and change in aerosol fluorescence intensity as a result of non-thermal plasma exposure. It has been established that the applied voltage is one of the principal variables for changing non-thermal plasma discharge power, generally leading to more efficacious chemical reactions or catalysis [35] or, in the case of pathogens, greater degrees of disinfection, including for airborne pathogens [15]. Sweeping applied voltage from 12 kV to 20 kV led to monotonic increases in viral aerosol inactivation (Table 2) as well as monotonic decreases in viral aerosol fluorescence intensity after plasma exposure. Regression analysis of the change in infectivity versus the difference in median fluorescence intensity yields an R^2 of 0.56. However, the quantitative relationship between the two variables still needs further investigation. Figure 6 presents a

Table 2 Plasma discharge power, aerosol fluorescence intensity level, and plasma-induced MS2 inactivation levels from different plasma peak-to-peak voltages. For the fluorescence intensity and virus infectivity, any row is statistically different from the plasma-off control (0 kV) with a p-value less than 0.05

Non-thermal plasma metrics		Aerosol fluorescence metrics		Change in aerosol metrics from NTP treatment	
NTP peak-to-peak voltage (kV)	NTP discharge power (W)	Median fluorescence intensity at outlet	Fluorescence intensity ratio compared to 0 kV control (%)	Change in fluorescence intensity difference	Change in virus infectivity (log PFU/mL*)*
0	0	$4.16(\pm 1.92)\times 10^6$	100	*0* (2.1×10^5 for the plasma-off control; subtracted from calculation; the same hereinafter)	*0* (0.21, determined by control tests and subtracted from calculation; the same hereinafter)
12	1.84	$3.91(\pm 1.86)\times 10^6$	94.10	2.4×10^5	0.23
16	2.44	$3.88(\pm 2.03)\times 10^6$	93.28	2.7×10^5	0.45
20	3.59	$3.87(\pm 2.07)\times 10^6$	93.11	2.9×10^5	1.03

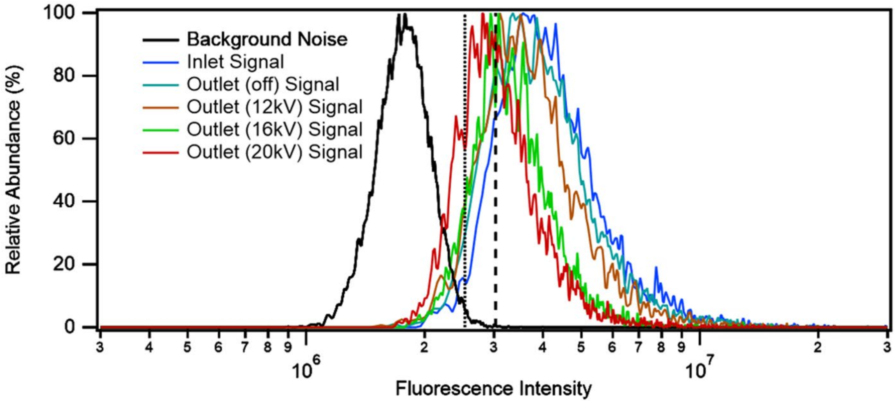

Fig. 6 Fluorescence intensity distributions of particles produced by aerosolizing MS2 stock, sampled at inlet (blue) and outlet (all other colors) of non-thermal plasma reactor as a function of applied peak-to-peak voltage (12 to 20 kV)

series of fluorescence intensity distribution curves for MS2-containing aerosols sampled at the NTP reactor outlet, curves which shift gradually towards lower intensity values (left) as the applied NTP voltage increases.

Numerous studies have previously explored non-thermal plasma treatment of viruses held in batch liquid solutions. However, compared to the present study, the differences in both timescales of plasma exposure (tens of seconds to minutes for batch liquids versus milliseconds for aerosols) and media (aqueous solution versus air) make the present findings more directly relevant to the subject of air disinfection by direct NTP treatment. Furthermore, the present results correlating viral aerosol fluorescence intensity with viral aerosol infectivity represent proof-of-principle for developing real-time, in situ optical monitoring of air flows before and after air disinfection treatment, which would allow much more rapid parametric performance evaluations and design iterations of such technologies than could be done by conducting microbiological assays at each test condition. However, a detailed causal mechanism is yet to be determined for the correlation between change in aerosol fluorescence intensity and loss of viral aerosol infectivity; in particular, yet to be determined are the thresholds below which minor losses of viral aerosol infectivity cannot be definitively correlated with change in aerosol fluorescence, or minor changes in aerosol fluorescence do not correspond to loss of viral aerosol infectivity. It has been hypothesized that proteins may play a central role: protein damage caused by NTP exposure could diminish the propensity of natural bioaerosol fluorophores (tryptophan and other amino acids) to fluoresce, as well as lead to loss of virus infectivity. In the *Supplemental Information* section, preliminary results exploring changes in protein structures resulting from viral aerosol exposure to NTP are presented.

In the present study, a correlation between the virus infectivity and its fluorescence intensity (or an anticorrelation between plasma-induced inactivation and fluorescence intensity) is observed. Specifically, viral aerosols exposed to non-thermal plasmas of increasing power were found to have both decreasing infectivity based on quantitative plaque assays and decreasing fluorescence intensity.

Concluding Remarks

This study presents the observed correlations between the change in UV fluorescence and the loss of infectivity for viral aerosols after non-thermal plasma treatment. A statistically significant weaker MS2 aerosol fluorescence is observed after exposure to non-thermal plasma, accompanied by loss of viral infectivity. Viral aerosol fluorescence intensity becomes weaker, and viral infectivity is increasingly reduced with more intense plasma treatment in terms of plasma discharge power. As such, aerosol fluorescence can be utilized to serve as a supporting diagnostic method for studying virus infectivity and virus protein structure. WIBS measurements of aerosol fluorescence have the potential to provide quick, real-time information for possible bioaerosol infectivity change and protein structural change. The findings and detection methods involving fluorescence are not limited to viruses and NTP inactivation, but could potentially also be applied to bacteria or allergens, whose fluorescence is also studied [36], and their corresponding treatment methods.

Limitations of this Work

The demonstrated correlation between viral aerosol fluorescence intensity changes and viral aerosol infectivity changes should be balanced against the limitations of the study. While non-thermal plasma exposure has been found to cause detectable changes in aerosolized MS2 virus fluorescence, it is unknown whether the same is true of other air disinfection methods. It is possible that other air disinfection methods are effective without causing a detectable change in aerosol fluorescence, or that other methods induce changes in aerosol fluorescence despite minimal effectiveness in air disinfection. The relationship between aerosol fluorescence and infectivity has yet to be established for other types of aerosolized pathogens, such as bacteria or enveloped viruses. From a practical standpoint, extending this work to other pathogens would likely require careful attention to potential interference from transport media components that differ from those used in the present study. The nature of the dispersion of virions and media components within the dried particles and their potential impact on the virus disinfection and fluorescence detection process have not been studied. In addition, the reported fluorescence changes are observed on the basis of the relative abundance of particles, which does not exclude the possibility of disproportionate fluorescence change among particles with higher fluorescence after NTP exposure, instead of a uniform decrease among all particles. Extensive work would be required to establish a quantitative relationship or a numerical model between the level of change in fluorescence intensities and in virus infectivity.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11090-026-10648-6.

Acknowledgements The authors would like to give their acknowledgments to Prof. Krista Wigginton, Dr. Anubhav Dwivedi, Dr. Peter Arts, Hunter Richards, Emily Hong, and Steve Donajkowski for their help with the execution of this study and to express gratitude to Dr. Dagen Hughes, Kristie Stowers, and Droplet Measurement Technologies for their assistance with WIBS instrument. The authors are sincerely grateful for the expertise and suggestions provided by Dr. Ke Zhang.

Author Contributions Z.M. and H.L.C. conceptualized the research; Z.M. performed the research; Z.M. analyzed the data; Z.M. and H.L.C. wrote the paper; H.L.C. provided resources; H.L.C. supervised the study; H.L.C. acquired funding. The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Funding The authors do not have any fundings to declare for this study.

Data Availability If accepted and published, the original dataset will be made available to all readers by request or from the dataset (https://orcid.org/0009-0006-3403-5684), shared via the University of Michigan online data repository Deep Blue Data (https://deepblue.lib.umich.edu/data). They can be found and accessed via the following: Ma, Z. (2025). Dataset for Fluorescence Measurements of MS2 Aerosol with Wideband Integrated Bioaerosol Sensor [Data set], University of Michigan - Deep Blue Data. https://doi.org/10.7302/tey0-ps17.

Declarations

Competing interests Non-financial interests: Author Herek L. Clack is co-founder and chief scientific officer at Taza Aya, Inc. Financial interests: The authors declare they have no financial interests.

References

1. Douwes J et al (2003) Bioaerosol health effects and exposure assessment: progress and prospects. Ann Occup Hyg 47:187

2. Herfst S et al (2014) *Avian influenza virus transmission to mammals.* Influenza Pathogenesis and Control-Volume I, : pp. 137–155
3. Imai M et al (2013) Transmission of influenza A/H5N1 viruses in mammals. Virus Res 178(1):15–20
4. Bai L et al (2021) Coinfection with influenza A virus enhances SARS-CoV-2 infectivity. Cell Res 31(4):395–403
5. What were the historical reasons for the resistance to recognizing airborne transmission during the COVID-19 pandemic? Indoor Air 32(8):e13070
6. Fernandez MO et al (2019) Assessing the airborne survival of bacteria in populations of aerosol droplets with a novel technology. J Royal Soc Interface 16(150):20180779
7. Lednicky J et al (2016) Highly efficient collection of infectious pandemic influenza H1N1 virus (2009) through laminar-flow water based condensation. Aerosol Sci Technol 50(7):i–iv
8. Mainelis G (2020) Bioaerosol sampling: classical approaches, advances, and perspectives. Aerosol Sci Technol 54(5):496–519
9. Schneider RK (2001) Designing clean room HVAC systems. ASHRAE J 43(8):39
10. Reisman RE et al (1990) A double-blind study of the effectiveness of a high-efficiency particulate air (HEPA) filter in the treatment of patients with perennial allergic rhinitis and asthma. J Allergy Clin Immunol 85(6):1050–1057
11. Mittal H et al (2011) Survival of microorganisms on HEPA filters. Appl Biosaf 16(3):163–166
12. Miaskiewicz-Peska E, Lebkowska M (2012) Comparison of aerosol and bioaerosol collection on air filters. Aerobiologia 28:185–193
13. Gallagher MJ et al (2007) Rapid inactivation of airborne bacteria using atmospheric pressure dielectric barrier grating discharge. IEEE Trans Plasma Sci 35(5):1501–1510
14. Liang Y et al (2012) Rapid inactivation of biological species in the air using atmospheric pressure nonthermal plasma, vol 46. Environmental Science & Technology, pp 3360–3368. 6
15. Wu Y et al (2015) MS2 virus inactivation by Atmospheric-Pressure cold plasma using different gas carriers and power levels. Appl Environ Microbiol 81(3):996–1002
16. Xia T et al (2020) Inactivation of airborne Porcine reproductive and respiratory syndrome virus (PRRSv) by a packed bed dielectric barrier discharge non-thermal plasma. J Hazard Mater 393:122266
17. Ma Z, Dwivedi AK, Clack HL (2024) Effects of chemically-reductive trace gas contaminants on nonthermal plasma inactivation of an airborne virus. Sci Total Environ 939:173447
18. Xia T et al (2019) Inactivation of airborne viruses using a packed bed non-thermal plasma reactor. J Phys D 52(25):255201
19. Nayak G et al (2020) Rapid inactivation of airborne Porcine reproductive and respiratory syndrome virus using an atmospheric pressure air plasma. Plasma Processes Polym 17(10):1900269
20. Zhang J et al (2020) Study of viral filtration performance of residential HVAC filters. ASHRAE J 62(8):26–32
21. Tseng C-C, Li C-S (2005) Inactivation of Virus-Containing aerosols by ultraviolet germicidal irradiation. Aerosol Sci Technol 39(12):1136–1142
22. Johnson W et al (2004) Pathogenic and humoral immune responses to Porcine reproductive and respiratory syndrome virus (PRRSV) are related to viral load in acute infection. Vet Immunol Immunopathol 102(3):233–247
23. Cornax R et al (1990) Application of direct plaque assay for detection and enumeration of bacteriophages of bacteroides fragilis from contaminated-water samples. Appl Environ Microbiol 56(10):3170–3173
24. Cormier J, Janes M (2014) A double layer plaque assay using spread plate technique for enumeration of bacteriophage MS2. J Virol Methods 196:86–92
25. Youil R et al (2004) Comparative study of influenza virus replication in Vero and MDCK cell lines. J Virol Methods 120(1):23–31
26. Swanson BE, Huffman JA (2020) Pollen clustering strategies using a newly developed single-particle fluorescence spectrometer. Aerosol Sci Technol 54(4):426–445
27. Stanley WR et al (2011) Continuous bioaerosol monitoring in a tropical environment using a UV fluorescence particle spectrometer. Atmospheric Sci Lett 12(2):195–199
28. Pöhlker C, Huffman JA, Pöschl U (2012) Autofluorescence of atmospheric bioaerosols – fluorescent biomolecules and potential interferences. Atmos Meas Tech 5(1):37–71
29. Toprak E, Schnaiter M (2013) Fluorescent biological aerosol particles measured with the waveband integrated bioaerosol sensor WIBS-4: laboratory tests combined with a one year field study. Atmos Chem Phys 13(1):225–243
30. Savage NJ et al (2017) Systematic characterization and fluorescence threshold strategies for the wideband integrated bioaerosol sensor (WIBS) using size-resolved biological and interfering particles. Atmos Meas Tech 10(11):4279–4302
31. Saari S, Reponen T, Keskinen J (2014) Performance of two fluorescence-based real-time bioaerosol detectors: BioScout vs. UVAPS. Aerosol Sci Technol 48(4):371–378

32. Kanaani H et al (2008) Performance of UVAPS with respect to detection of airborne fungi. J Aerosol Sci 39(2):175–189
33. Mesquita M et al (2010) Optimal Preparation and purification of PRD1-like bacteriophages for use in environmental fate and transport studies. Water Res 44(4):1114–1125
34. Wick C, McCubbin P (1999) Characterization of purified MS2 bacteriophage by the physical counting methodology used in the integrated virus detection system (IVDS). Toxicol Methods 9(4):245–252
35. Bahri M et al (2016) Impact of design parameters on the performance of non-thermal plasma air purification system. Chem Eng J 302:204–212
36. Jung JH, Lee JE (2013) Variation in the fluorescence intensity of thermally-exposed bacterial bioaerosols. J Aerosol Sci 65:101–110